Assignment 3

Total points 200, passing marks 130.

1. Develop a problem statement (real world and machine learning) (10)
    a. This is one of the most important skills that a Machine Learning Engineer/Scientist should have. Select a dataset and frame a machine learning problem and then connect this machine learning problem to the real world scenario.
2. Implement the preprocessing and justify the preprocessing steps (you can reuse from assignment 1)
3. Extract features and justify the methods used (20, without justification 10)
4. Select features or use a dimensionality reduction mechanism and justify the methods used (20, without justification 10). If you are not using either of these techniques, justify why not.
5. Implement three out of the following algorithms and justify the choice (30, without justification 10).
    a. K-nearest neighbours
    b. Regression (linear or polynomial)
    c. Additive model
    d. Decision trees
    e. Naïve Bayes
    f. SVM with kernels
    g. Neural Network
6. Compare the performance of the three algorithms with respect to your problem, explain the results. (20, without explanation 10)
7. Implement boosting and bagging with your choice of base models and explain all the steps (20). Use different algorithms for boosting and bagging and justify your choices.
8. Compare the performance of the algorithms (basic VS boosting VS bagging VS transfer) with respect to your machine learning problem and explain the results. (20, without explanation 10)
9. Implement a Convolution Neural Network or a deep Neural Network with hyperparameter tuning (for this you can directly use the data after the preprocessing) (30). Justify the hyperparameters tuned.
10. Compare and explain the results in terms of the complexity, the computation time and the performance of the classification algorithms. This has to also take into account the machine learning problem and the real world problem. (30)